



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE G549

Course Title : Rural Road Technology

Instructor-in-charge : Dr. Vijay Bajirao Kakade

[Team of Instructors] : -

1. Course description:

Network planning, accessibility and mobility; road alignment and survey; geometric design: cross sectional elements, sight distance, horizontal and vertical alignments; road materials and use of marginal materials; pavement design, drainage, culverts and small bridges; construction and specifications; quality control in construction; pavement failures; maintenance; preparation of detailed project report (DPR); community participation in planning, design, construction and management.

2. Scope and Objective:

Rural roads are the key components for the development of rural community. In this regard, the Pradhan Mantri Gram Sadak Yojana (PMGSY) was launched by the Govt. of India to provide connectivity to unconnected Habitations as part of a poverty reduction strategy. Subsequently, the various strategies have been adopted in planning and construction of rural roads with the aid of technologically advanced construction practices. The scope of the course is to know the various aspects rural roads construction and the various processes associated with the construction. In the recent past, special emphasis is on the use of green technology for road construction in many states. Rural road technology is fast evolving at an urban pace. The course aims to cover various technological advancements that have taken place not only in India, but also on the global stage.

3. Course learning outcomes:

- a. .To perform the geometric design of rural roads
- b. .To design the flexible and rigid pavement
- c. .To prepare the detailed project report

4. Text books and references:

TB1 : IRC:SP 20 Rural Roads Manual.



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

- R1** : Quality Assurance Handbook for Rural Roads - published by NRRDA, Volume I & II
R2 : IRC:SP 72 -2015, Guidelines for the design of flexible pavements for low volume roads
R3 : IRC: SP 62 – 2014, Guidelines for design and construction of cement concrete pavements for Low volume roads
R4 : PMGSY I, II and III guidelines
R5 : Specifications for Roads and Bridge Works, Fifth Revision, Ministry of Road Transport and Highways, 2013.
R6 : Guide on Managing Maintenance of Rural Roads in India, NRRDA publication
R7 : Specifications for rural roads, First Revision, Ministry of Rural Development, 2014
R8 : Other relevant IRC, BIS codes and peer reviewed journal articles

5. Method of conduct of the course: Power point presentation

6. Course Plan:

Week no.	Module	Lect. no.	Reference	Learning Outcome
1.	Planning of Rural Roads – Introduction to rural road technology; classification of roads; road development plans; Low connectivity	1-3	T1, R4, R8	To understand the importance of proper planning of rural roads and the models adopted to achieve it. To learn the salient features of PMGSY and importance of DPR in rural road
2.	Planning of Rural Roads – network planning Models, accessibility and mobility; PMGSY history, salient features, NRRDA; Introduction to Detailed Project Report (DPR)	4-6	T1, R4, R8	To understand the importance of proper planning of rural roads and the models adopted to achieve it. To learn the salient features of PMGSY and importance of DPR in rural road
3.	Geometric Design of Rural Roads – Road alignment and survey; geometric design : cross section elements, Sight distances	7-9	T1, R7, R8	To understand the geometric design
4.	Geometric Design of Rural Roads – Horizontal alignment, Vertical alignment, safety aspects, Incorporation of geometric design in DPR preparation	10-12	T1, R7, R8	To understand the geometric design



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

5.	Materials for rural roads – Characteristics of pavement materials including soil and marginal materials; Stabilization of soil; Locally available materials; cost and durability aspects; Material specifications in IRC codes	13-15	T1, R5, R7, R8	To understand the materials adopted and selection criteria in the construction of rural roads
6.	Design of Rural Roads – Design parameters, Pavement components, Design of flexible Pavements	16-18	T1, R2, R3, R8	To design rural roads (Flexible) based on the current specifications followed by PMGSY
7.	Design of Rural Roads – Semi – Rigid and Rigid pavements; Drainage and shoulders, types and criteria for road drainage; system of drainage; Incorporation of design in DPR.	19-21	T1, R2, R3, R8	To design rural roads (rigid) based on the current specifications followed by PMGSY. To learn the importance, types and selection of drainage.
8.	Detailed Project Report – Preparation of DPR; Important aspects of DPR; Surveys and investigation; Representations of Designs & Drawings in DPR; Role of STA/PTA in evaluating the DPR; Aspects in DPR scrutiny; Community participation in planning, construction, maintenance	22-24	T1, R1-R8	Formats and Proforma of DPR, Road safety, Cost and rate analysis in DPR
9.	Construction, failures and maintenance of rural roads - Construction of rural roads and quality control	25-27	T1, R1, R5, R6, R7	To assess the quality control in construction



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

10.	Construction, failures and maintenance of rural roads - Failures and distress in flexible pavements	28-30	T1, R1, R5, R6, R7	To learn to assess the failure and maintenance methods adopted.
11.	Construction, failures and maintenance of rural roads - Failures and distress in rigid pavements	31-33	T1, R1, R5, R6, R7	To learn to assess the failure and maintenance methods adopted.
12.	New Technologies - Use of innovative and cost effective construction materials	34-36	T1, R4, R8	To understand the new technologies adopted in the construction of rural roads and their role in energy reduction.
13.	New Technologies - Cell filled concrete; Use of fly ash, waste materials; Plastic wastes	37-39	T1, R4, R8	To understand the new technologies adopted in the construction of rural roads and their role in energy reduction.
14.	New Technologies - Emulsion based cold mix technology; Less energy intensive technology and materials	40-42	T1, R4, R8	To understand the new technologies adopted in the construction of rural roads and their role in energy reduction.

7. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quiz	21%	30 min	Quiz 1: 3/9/26 Quiz 2: 1/10/26 Quiz 3: 19/11/26	All quizzes are mandatory. Closed book
DPR Checking	19%	-	TBA	Open book
Mid Semester	25%	90 min	07/10/2026	Closed book
Comprehensive	35%	180 min	08/12/2026	Open book



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

Criterion for NC: If the student absents herself/ himself even in one of the evaluative components (listed in the evaluation scheme) entirely, his / her performance shall be reported as NC (Not Cleared) regardless of his/ her final total score in the course (See Clause 4.19 of BITS Academic Regulations). Students gaining less than 10% overall, may be awarded NC grade.

8. Chamber consultation hours: Tuesday 11 am – 12 pm

9. Makeup Policy:

- Make up will be granted only on genuine reasons and will be considered on a case by case basis. However, prior permission is a must. It should follow the official request procedure. No other mod of request shall be entertained.
- For medical cases, a certificate from the concerned physician from the Medical Centre must be produced. (Subjected to Academic regulations)
- Make up policy is only applicable to Mid – Semester test and Comprehensive Examination. It is not applicable to tests, assignment and term paper.

Dr. Vijay Kakade

Instructor-in-charge

Course No. CE G549